TTEC101 Assignment 1

Programming Foundations & Reproducible Workflow

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TTEC101 Assignment 1

Student Data Analysis and Version Control Workflow

1. Introduction

This assignment focuses on developing foundational skills in command-line operations, version control using Git, and basic data analysis using R. These tools are essential in modern data analytics workflows as they enable efficient project organisation, reproducibility, and collaboration.

The objective of this project was to create a structured data analysis environment, manage files using Git, and perform exploratory analysis on a student dataset. The assignment also emphasises documenting the workflow clearly using Markdown and reflecting on the learning experience.

2. Dataset Description

The dataset used in this project contains information about students and their academic performance. It includes several variables that help analyse how different factors influence final grades.

Key variables include:

- **Age:** The age of each student
- **Gender:** Male or Female
- **Study Hours:** Number of hours spent studying
- **Attendance Rate:** Percentage of classes attended
- **Assignment Score:** Marks obtained in coursework
- **Final Exam Score:** Marks obtained in the final exam
- **Final Grade:** Overall performance score

This dataset is useful for understanding patterns in student performance and identifying which factors contribute most to academic success.

3. Methodology

3.1 Project Setup Using Command Line

The project was created using the command-line interface. A main directory named **ttec101-assignment1** was created to organise all files related to the assignment. Within this directory, three subfolders were created:

- **data** – to store the dataset
- **scripts** – to store R scripts
- **docs** – to store documentation files

This structure ensures that files are well organised and easy to manage. Using the terminal to create and navigate directories also improves efficiency and familiarity with development tools.

Figure 1: Command-line folder creation and structure

```
CLANGARM64:/c/Users/podee/ttec101-assignment1

podee@Dipma CLANGARM64 /
$ cd ~

podee@Dipma CLANGARM64 ~
$ mkdir ttec101-assignment1
mkdir: cannot create directory 'ttec101-assignment1': File exists

podee@Dipma CLANGARM64 ~
$ cd ttec101-assignment1

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ mkdir data scripts docs
mkdir: cannot create directory 'data': File exists
mkdir: cannot create directory 'scripts': File exists
mkdir: cannot create directory 'docs': File exists

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ ls
analysis.R  data/  docs/  scripts/

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$
```

GitHub Repository:
https://github.com/dipma43-cell/ttec101-assignment1

The project was linked to a remote GitHub repository to support version control, and the workflow is demonstrated through the recorded commits and screenshots.

3.2 Version Control Using Git

Git was used to track changes throughout the project. The repository was initialised using the git init command. Files were then added and committed at different stages of the project.

Commits included:

- Initial project setup
- Adding the dataset
- Creating the R analysis script
- Writing the Markdown report
- Final updates

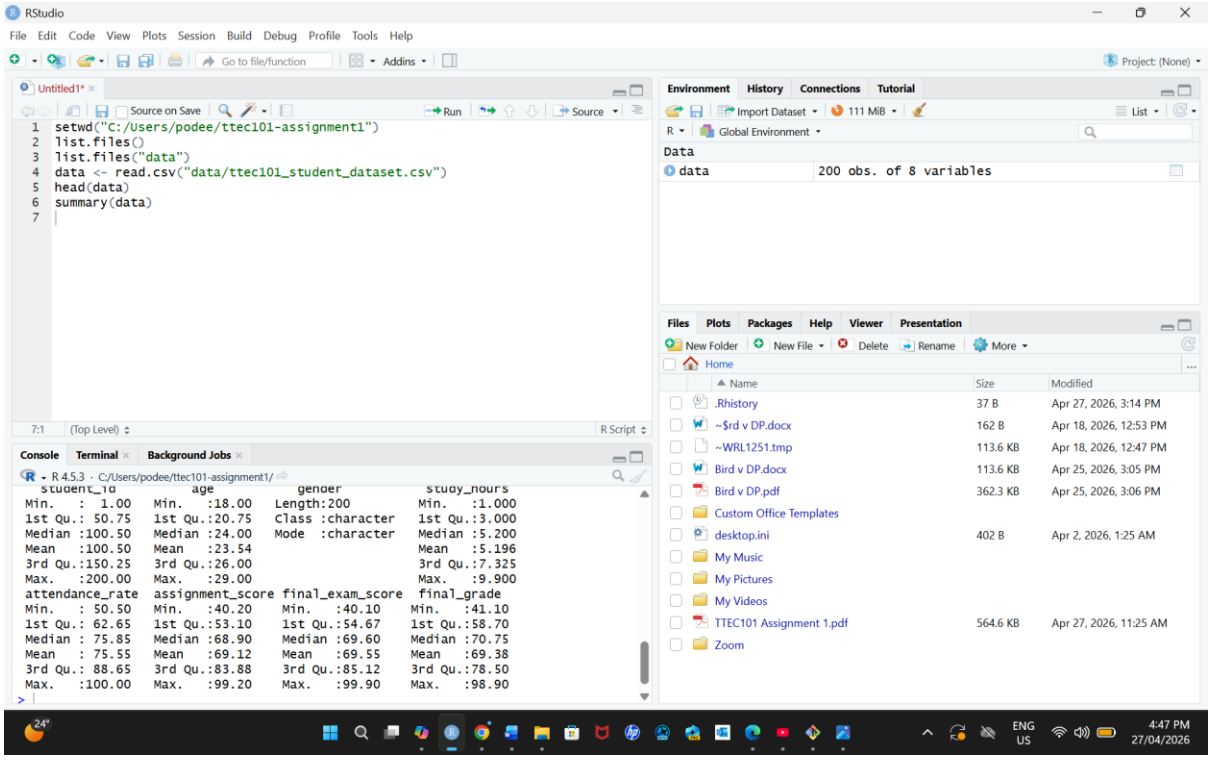
Each commit represents a checkpoint, allowing changes to be tracked and managed efficiently. This demonstrates the importance of version control in maintaining a clean and reproducible workflow.

Figure 2: Git repository initialization and commit history

```
podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ ls data
'ttec101_student_dataset (3).csv'

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ mv "data/ttec101_student_dataset (3).csv" data/ttec101_student_dataset.csv

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ ls data
ttec101_student_dataset.csv

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ git init
git add .
git commit -m "final assignment"
Reinitialized existing Git repository in C:/Users/podee/ttec101-assignment1/.git/
warning: in the working copy of '.Rhistory', LF will be replaced by CRLF the next time Git touches it
[main 1cfce93] final assignment
 2 files changed, 14 insertions(+)
 create mode 100644 .Rhistory
 rename data/{ttec101_student_dataset (3).csv => ttec101_student_dataset.csv} (100%)

podee@Dipma CLANGARM64 ~/ttec101-assignment1 (main)
$ |
```

3.3 Data Analysis Using R

An R script was created to analyse the dataset. The dataset was loaded using the read.csv() function, and basic exploration was performed using head() and summary().



The script used:

data <- read.csv("data/ttec101_student_dataset (3).csv")

head(data)

summary(data)

This allowed for quick inspection of the data and understanding of key statistics such as averages, ranges, and distributions.

4. Results

The dataset contains 200 students, with an average age of approximately 23.5 years. The distribution of academic performance shows that most students fall within a mid-range performance level.

Key findings include:

- The average study time is approximately 5.2 hours
- The average attendance rate is around 75.5%
- Assignment and final exam scores both average close to 69
- The average final grade is approximately 69.4

The most significant insight is that final exam scores have the strongest relationship with final grades, indicating that exam performance plays a major role in determining overall results.

Assignment scores also show a moderate relationship with final grades, suggesting that continuous assessment contributes meaningfully to performance.

However, study hours show only a weak relationship with final grades, indicating that simply studying longer does not necessarily guarantee better results. Similarly, attendance rate shows minimal impact on final grades, suggesting that attendance alone is not a strong predictor of success in this dataset.

Overall, the results suggest that assessment performance (especially exams) is the most important factor in student success.

5. Discussion

The findings from this analysis highlight the importance of performance-based assessment in academic outcomes. Final exam scores have a very strong influence on final grades, suggesting that exams are heavily weighted in determining overall performance.

The moderate relationship between assignment scores and final grades indicates that coursework also plays an important role. This reflects a balanced evaluation system where both exams and assignments contribute to student success.

Interestingly, study hours and attendance rate do not show strong relationships with final grades. This suggests that factors such as study quality, understanding of material, and exam preparation strategies may be more important than time spent studying or simply attending classes.

One limitation of this dataset is that it does not include qualitative variables such as learning styles or teaching methods, which may also influence performance.

6. Reflection

This assignment provided valuable experience in using Git and version control systems. Git helps track changes in files and allows different stages of a project to be saved through commits, making it easier to manage projects and avoid losing work.

Reproducibility is an important concept in data analytics because it ensures that results can be recreated by others. By organising files properly and using scripts, the analysis process becomes transparent and repeatable.

One challenge faced during this assignment was understanding how to use Git commands correctly, particularly committing and managing files. There were also initial difficulties with

command-line operations. However, through practice and repetition, these challenges were overcome.

Overall, this assignment improved confidence in using the command line, Git, and R for data analysis, while also highlighting the importance of organisation and documentation in any data-related project.

7. Conclusion

In conclusion, this assignment successfully demonstrated the integration of command-line tools, version control, and data analysis techniques. The use of Git allowed for efficient tracking of project progress, while R enabled basic exploration of the dataset.

The analysis revealed that exam performance is the most significant factor influencing student grades, followed by assignment scores. The project also emphasised the importance of reproducibility and structured workflows.

These skills are essential for future work in data analytics and will be valuable in both academic and professional contexts. This project demonstrates the importance of reproducibility and structured workflows in modern data analysis practices.

Note: Figures included in this report provide evidence of folder structure, Git commits, dataset integration, and overall project workflow.